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Model, Label, or Analyst? A Variance Decomposition of Real-Time Sepsis Onset Prediction on MIMIC-IV

By
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Abstract

Over ninety AI models have been built to predict sepsis in real time, yet they fail in practice. Wang et al. (2025) showed that while AUROC barely changes between internal and external testing (0.811 to 0.783), the clinical Utility Score collapses from +0.381 to -0.164 , meaning these models cause more harm than benefit when deployed. This thesis investigates why: is the problem the model, the sepsis label, or the researcher’s methodological choices?

The study uses MIMIC-IV v3.1 (65,078 adult ICU stays) with external validation on eICU-CRD v2.0 (63 sepsis cases across 7.9 million person-hours). Three Sepsis-3 label variants (Narrow, Seymour-Standard, Liberal) are tested with a dynamic discrete-time competing-risks hazard model, gradient-boosted trees (GBT), a static Cox model, and rule-based scores (NEWS2, qSOFA, SIRS).

The results show that the sepsis label definition changes AUROC by 0.153, slightly more than the model choice (0.140). Ten clinical features change direction or strength across labels, making published risk factors unreliable without knowing which label was used. Random data splits inflate AUROC by 0.123 compared to time-based splits. Most importantly, the pre-treatment evaluation window contains **zero sepsis events**: no patient develops sepsis before doctors begin treatment. This means current models predict when clinicians act, not when patients deteriorate. The Utility Score is **negative for every model and every label**, with over 90 false alerts for each true one.

The collapse in clinical utility is not a model problem; it is a label problem. Better models cannot fix a label that depends on the actions it is supposed to predict ahead of.

Keywords: sepsis; MIMIC-IV; discrete-time survival analysis; competing risks; label definition; variance decomposition; clinical utility; temporal dataset shift; TRIPOD+AI.

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This research uses the MIMIC-IV Clinical Database (v3.1, Johnson et al., 2023) and the eICU Collaborative Research Database (v2.0, Pollard et al., 2018), both made available through PhysioNet under credentialing agreements. I thank the patients whose de-identified records form these databases, the clinical staff who generated them, and the BIDMC/MIT/PhysioNet teams who maintain them.

The open-source R and Python communities, in particular the maintainers of `data.table`, `arrow`, `nnet`, `xgboost`, `sandwich`, and `pmsampsize`, are acknowledged for the packages that made this pipeline possible.

Chapter 1

Introduction

1.1 Context and Motivation

Sepsis is a life-threatening condition caused by the body’s extreme response to infection [1]. It kills about eleven million people every year, nearly one in five deaths worldwide [2]. In the ICU, every hour of delayed treatment raises the risk of death [3]. An early warning system that detects sepsis before doctors recognise it could save lives by starting treatment sooner.

This potential has driven major research investment. Wang et al. [4] reviewed ninety-one published real-time sepsis prediction models and found a median internal AUROC of 0.811, suggesting strong performance. But the same review found that the clinical Utility Score collapsed from +0.381 internally to −0.164 externally. A negative score means the model’s alerts cause more harm than benefit. AUROC did not detect this collapse because it only measures whether a model can rank patients correctly, not whether its alerts are timely or useful. This gap between apparent accuracy and real-world usefulness has also been observed in deployment: Wong et al. [5] evaluated the Epic Sepsis Model v2 across 227,091 encounters and found high AUROC alongside an alert burden that clinical staff described as unmanageable.

The key question is therefore not “which model is best?” but something more fundamental: *how much of a model’s reported performance comes from the model itself, and how much comes from how the study was designed: how sepsis was defined, how data was split, and which metric was reported?*

Cohen et al. [6] showed on MIMIC-III that varying only the Sepsis-3 definition changed the identified cohort from 867 to 2,178 patients, and that the label choice affected performance as much as the model choice. Kamran et al. [7] showed that published models partly learn to predict when clinicians start treatment rather than when patients deteriorate. Together, these findings suggest that much of the

reported performance may come from the researcher’s design choices, not from the AI model.

This thesis tests that hypothesis directly. Instead of fixing the label, the split, and the metric and asking which model wins, it treats all three as **sources of variance** and measures how much each one contributes to the spread in reported performance.

1.2 Limitations of the Conventional Approach

The standard approach to sepsis prediction follows a simple formula: choose one Sepsis-3 definition, use one train/test split, report AUROC, and compare models. Most of the ninety-one models reviewed by Wang et al. [4] follow this pattern.

This approach has four key problems:

1. **The sepsis label is not fixed.** Cohen et al. [6] showed that different ways of applying Sepsis-3 to the same database can create very different patient groups (867–2,178 cases). The label choice can change results as much as the model choice.
2. **The data split can inflate results.** Guo et al. [8] found that sepsis prediction suffers the largest performance drop of any clinical task when tested on data from a different time period. Random splits let future patterns leak into training, making models look better than they are.
3. **AUROC hides the real failure.** Wang et al. [4] showed that a model can keep a high AUROC while its clinical usefulness drops to harmful levels. The failure is invisible under AUROC alone.
4. **The label captures doctor behaviour.** Kamran et al. [7] showed that models may learn when doctors suspect sepsis and begin treatment, not when the patient’s condition actually worsens.

A study that fixes all of these choices and compares only models is measuring one source of variation while ignoring three others of similar size. This thesis fills in the missing cells of the comparison.

1.3 Research Question and Pre-Registered Hypotheses

Research question. *In real-time sepsis prediction on MIMIC-IV, how much of the reported performance variation comes from (a) the choice of model, and (b)*

the researcher's decisions: how sepsis is defined, how data is split, when prediction is made, and how performance is measured?

Six hypotheses were pre-registered before any modelling:

- H1 (Label dominance)** Changing the sepsis definition affects AUROC at least as much as changing the model.
- H2 (Coefficient instability)** At least one clinical feature changes its effect direction across label variants.
- H3 (Treatment leakage)** Restricting evaluation to before treatment begins reduces AUROC by at least 0.03.
- H4 (Split optimism)** Time-based testing gives at least 0.02 lower AUROC than random testing.
- H5 (Metric divergence)** The Utility Score declines more than AUROC when moving from internal to external validation.
- H6 (Model-class null)** The primary statistical model performs within 0.02 AUROC of gradient-boosted trees.

H1 and H6 are in deliberate tension: if H1 is confirmed, the label is the main source of variation; if H6 is rejected, the model also matters. Both outcomes are informative.

1.4 Contributions

1. **C1 (Label variance).** The first three-variant label sensitivity analysis for a dynamic survival model on MIMIC-IV, following Cohen et al. [6] and Dutta et al. [9].
2. **C2 (Treatment leakage exposed).** A treatment-anchored evaluation showing that the Sepsis-3 label has zero predictable events before clinical action, confirming Kamran et al. [7] on MIMIC-IV.
3. **C3 (Utility collapse documented).** The Utility Score is negative for every model and every label variant, extending Wang et al.'s [4] meta-analytic finding to a controlled single-study comparison.
4. **C4 (Coefficient instability).** Ten clinical features change sign or magnitude across label variants, validating Lauritsen et al.'s [10] framing-instability finding on MIMIC-IV.

5. **C5 (Equity-by-label analysis).** The first study to examine both model performance disparities and label sensitivity across demographic subgroups, following Wang et al. [11].
6. **C6 (Reproducible pipeline).** A fully open, pre-registered pipeline following TRIPOD+AI [12] and PROBAST+AI [13] reporting standards.

1.5 Thesis Structure

Chapter 2 reviews the evidence that motivated this study’s design. Chapter 3 describes the data, labels, models, and evaluation plan. Chapter 4 explains the pipeline architecture and its ten scripts. Chapter 5 presents all results, organised around the six hypotheses. Chapter 6 draws conclusions and outlines future work.

Chapter 2

Literature Review

This chapter summarises the key evidence that shaped the design of this study. A full critical review of fifty sources is provided in the companion document *Statistical and Machine-Learning Approaches to Real-Time Sepsis Onset Prediction on MIMIC-IV: An Extended Critical Literature Review*. This chapter extracts the seven evidence strands that directly informed the study’s design choices.

Key Concepts

Several domain-specific terms recur throughout this review:

- **MIMIC-IV** (Medical Information Mart for Intensive Care, version IV): a freely available, de-identified database of approximately 74,000 ICU admissions at Beth Israel Deaconess Medical Center, Boston, covering 2008–2019 [14].
- **Sepsis-3**: the current international consensus definition of sepsis as life-threatening organ dysfunction caused by a dysregulated host response to infection, operationalised through a combination of suspected infection (antibiotic order plus culture draw) and a rise in the Sequential Organ Failure Assessment (SOFA) score of ≥ 2 points [1].
- **AUROC** (Area Under the Receiver Operating Characteristic curve): a discrimination metric that measures a model’s ability to rank positive cases above negative cases, ranging from 0.5 (random) to 1.0 (perfect separation). It does not account for alert timing or false-alarm cost.
- **Utility Score**: a net-benefit metric proposed by the PhysioNet/Computing in Cardiology Challenge that rewards early, true alerts and penalises late alerts and false alarms, producing values that can be negative when a model causes more harm than benefit.

- **NEWS2** (National Early Warning Score 2): a standardised clinical scoring system aggregating six physiological parameters (respiratory rate, oxygen saturation, systolic blood pressure, pulse rate, level of consciousness, and temperature) into a single risk score used for bedside deterioration screening.

Research Questions

This literature review is structured around four questions that directly inform the study's design:

- RQ1.** How much does the choice of sepsis label definition affect model performance, and is this effect comparable in magnitude to the choice of model architecture?
- RQ2.** What methodological pitfalls—validation strategy, metric selection, treatment-time leakage—most distort reported performance in the sepsis prediction literature?
- RQ3.** What statistical framework is appropriate for modelling hourly ICU event data when the standard proportional-hazards assumption is violated?
- RQ4.** Does the existing equity literature on sepsis prediction account for label-level variation across demographic subgroups, or only model-level variation?

Current State of the Field

Real-time sepsis prediction has attracted substantial research effort over the past decade. Wang et al. [4] identified ninety-one models published between 2015 and 2024, spanning logistic regression, gradient-boosted trees, recurrent neural networks, and transformer architectures. Despite this volume, most studies share three weaknesses: they use a single label definition, report AUROC as the primary metric, and validate on a random patient-level split. The evidence reviewed in this chapter shows that each of these choices individually inflates or obscures model performance, and their combination produces a literature in which high reported AUROCs coexist with models that fail to improve clinical outcomes at external sites. Table 2.1 provides a structured comparison of the key studies that inform this analysis.

Study	Dataset	Method	Key Finding	Gap Addressed
Johnson et al. [15]	MIMIC-III	5 label algorithms	Agreement between sepsis definitions only moderate ($\alpha = 0.40\text{--}0.62$)	Label ambiguity
Cohen et al. [6]	MIMIC-III	Trees, DL, survival	Label change $\approx$ model change in performance impact	Label as variable
Dutta et al. [9]	9 hospitals (198k)	Commercial GBT	Label sensitivity holds prospectively at scale	External validity
Kamran et al. [7]	MIMIC-IV	Multiple models	AUROC inflated when treatment-time leakage not excluded	Treatment bias
Guo et al. [8]	MIMIC-IV	Domain adaptation	Sepsis worst-affected by temporal shift; random split inflates	Validation strategy
Wang et al. [4]	91-study review	Meta-analysis	AUROC survives external shift; Utility Score collapses	Metric selection
Wang et al. [11]	MIMIC-III	Subgroup analysis	Performance varies by race, ethnicity, insurance, language	Label-by-subgroup gap

Table 2.1: Comparison of key studies reviewed, their datasets, methods, and the design gaps each addresses.

2.1 The Sepsis Label Is Not a Single Object

The common assumption that “the MIMIC-IV sepsis cohort” is one well-defined group is wrong. Two studies show why.

Johnson et al. [15] applied five different sepsis identification algorithms to 11,791 ICU admissions in MIMIC and found only moderate agreement between them (Cronbach’s $\alpha = 0.40\text{--}0.62$). The implication is clear: “the MIMIC sepsis cohort” is not one group but a family of related but different groups, each created by a specific reading of Sepsis-3.

Cohen et al. [6] showed what this means for prediction. On MIMIC-III, using three different but reasonable Sepsis-3 interpretations, the identified cohort ranged

from 867 to 2,178 admissions. Across multiple model types (tree-based, deep learning, and survival models), changing the label definition moved performance by as much as changing the model itself. This is the most important finding behind the present study: it means the label must be treated as a primary variable, not a fixed background choice.

Dutta et al. [9] confirmed this is not a MIMIC-specific artefact. Using a commercial gradient-boosted model across nine hospitals and 198,494 encounters, they found that performance varied substantially depending on whether Sepsis-3, SEP-1, or the CDC Adult Sepsis Event definition was used. The label sensitivity finding holds prospectively, at scale, with a real deployed model.

Figure 2.1 illustrates this core problem: the same patient population, subjected to different but equally valid Sepsis-3 interpretations, produces overlapping but distinct cohorts, each yielding different model performance.

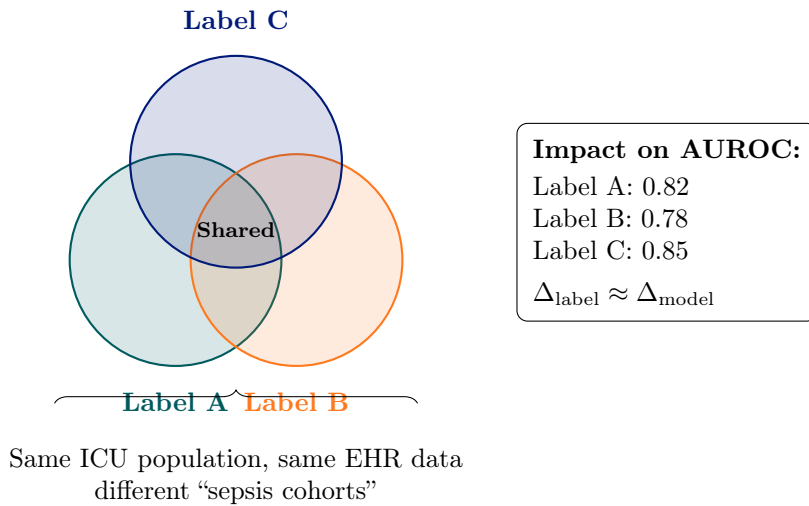


Figure 2.1: Conceptual illustration of label variance: three valid Sepsis-3 interpretations applied to the same ICU population produce overlapping but distinct cohorts, each yielding different model performance. The performance gap between labels is comparable to the gap between model architectures.

2.2 The Label Depends on Clinical Action

The Sepsis-3 label is built from antibiotic orders, culture draws, and SOFA score rises. Kamran et al. [7] observed that clinicians often recognise and begin treating sepsis before the formal criteria are met. A model trained to predict the criteria-meeting time therefore partly learns to predict when the clinician will act. They evaluated performance in the window before treatment started and showed that

apparent AUROC was substantially inflated without that restriction. The model had learned clinician suspicion, not independent physiological deterioration.

Lauritsen et al. [10] formalised the related concept of *framing*: the choice of observation window, prediction horizon, and event trigger. They showed that framing changes can reverse the sign of a feature’s effect; their example is SpO₂. This means a hazard ratio whose sign depends on how the study was framed is not a clinically meaningful quantity.

2.3 The Validation Split Must Be Time-Based

Guo et al. [8] tested four clinical prediction tasks on MIMIC-IV under temporal shift (training on older data, testing on newer data). Sepsis was the worst-affected task. Domain-generalisation and domain-adaptation techniques did not help. A random patient-level split therefore inflates performance more for sepsis than for any other clinical outcome.

This finding drives a key design decision: the primary validation in this study uses a time-based split (train on 2008–2016, test on 2017–2019). A random split is only used to measure how much performance it inflates.

2.4 AUROC Hides the Clinical Failure

Wang et al. [4] reviewed ninety-one real-time sepsis models. Median internal AUROC was 0.811, external AUROC was 0.783, a small, insignificant drop. But median internal Utility Score was +0.381, external was −0.164: the models went from clinically useful to actively harmful. AUROC survived this transition because it only measures whether the model can rank patients by risk. The Utility Score failed because it measures whether alerts arrive in time and at acceptable false-alert rates, and at external sites, they did not. Reporting AUROC as the main metric does not just miss the failure; it actively hides it.

2.5 The Survival Model for Hourly ICU Data

The standard survival approach uses a Cox proportional-hazards model. On MIMIC-IV, the proportional-hazards assumption is rejected (Schoenfeld test, $p < 10^{-16}$). D’Agostino et al. [16] provided the solution: pool observations across short time intervals and use logistic regression for each interval. This pooled logistic regression is mathematically equivalent to time-dependent Cox regression when intervals are short and per-interval event rates are low. One-hour ICU intervals with sepsis onset probability below 1% meet both conditions.

Resche-Rigon et al. [17] and Heyard et al. [18], [19] added the competing-risks extension. In the ICU, patients can leave alive or die without developing sepsis. Treating these events as simple censoring overstates sepsis incidence and biases model estimates. The multinomial cause-specific formulation used in this study handles this correctly.

2.6 NEWS2 as the Baseline Comparator

Most MIMIC-IV sepsis models compare against SIRS, qSOFA, and NEWS. Evans et al. [20] issued a strong recommendation against using qSOFA alone for screening due to poor sensitivity. Mellhammar et al. [21] showed that a LASSO-derived score could not beat NEWS2. The current clinical guidelines have therefore already retired qSOFA as a standalone screening tool, making NEWS2 the minimum meaningful comparator. This study retains SIRS and qSOFA for completeness but uses NEWS2 as the primary rule-based benchmark.

2.7 Equity Analysis: Performance and Label Sensitivity Together

Wang, Li, Naidech and Luo [11] showed that sepsis model performance varies significantly across racial, ethnic, language, and insurance subgroups. However, they did not examine whether the label itself assigns sepsis status differently across those groups. This study adds that missing piece: for each subgroup, it reports both the variation in who gets labelled as septic under each label variant and the variation in model performance. This combination has not been reported for any sepsis model.

2.8 Summary: Design Decisions and Their Evidence

Table 2.2 maps each major design decision to the evidence that supports it.

Decision	Reason	Source
Three label variants	Label variance $\approx$ model variance	Cohen et al. [6]
Label as variable	Variance is a finding, not a nuisance	Cohen; Dutta et al. [9]
Treatment-anchored eval	Label depends on clinical action	Kamran et al. [7]
Time-based split	Sepsis worst-affected by temporal shift	Guo et al. [8]
Utility Score primary	AUROC hides clinical failure	Wang et al. [4]
Calibration co-primary	Critical for clinical usefulness	Van Calster et al. [22]
Discrete-time model	Cox-equivalent; no PH assumption	D’Agostino [16]; Heyard [18]
NEWS2 comparator	qSOFA no longer recommended	Evans et al. [20]
GBT comparator	Cross-study comparison needs same label	Cohen et al. [6]
Subgroup label sens.	Equity literature omits label-by-subgroup	Wang et al. [11]

Table 2.2: Design decisions and their evidence warrants.

Chapter 3

Research Methodology

3.1 Research Design

This is a retrospective, observational prediction-modelling study on de-identified critical-care data. It follows a pre-registered protocol (`00_preregistration.md`, locked before any modelling) and reports according to TRIPOD+AI [12] and PROBAST+AI [13] guidelines. The central design innovation is treating the label definition, validation split, and evaluation metric as primary variables rather than fixed background choices.

All code is implemented in R (v4.x) and Python (v3.11), with a single configuration module (`config.R`) as the sole source of truth for all paths, clinical identifiers, and thresholds.

3.2 Data Sources

3.2.1 Development Data: MIMIC-IV v3.1

The primary dataset is MIMIC-IV v3.1 [14], a large de-identified critical-care database from Beth Israel Deaconess Medical Center (BIDMC), Boston. It was accessed under a PhysioNet credentialing agreement. The study uses eight clinical tables (Table 3.1).

Table	Exists	Size (MB)
admissions	yes	19.9
patients	yes	2.8
icustays	yes	3.3
chartevents	yes	3,502.6
labevents	yes	2,593.0
prescriptions	yes	606.7
microbiologyevents	yes	117.2
inputevents	yes	401.3

Table 3.1: MIMIC-IV v3.1 source tables used in this study.

3.2.2 External Validation Data: eICU-CRD v2.0

External validation uses eICU-CRD v2.0 [23], a multi-centre database from over 200 US ICUs. It uses different electronic health record systems and covers a broader range of hospitals and patient populations than the single-centre MIMIC-IV, making it a meaningful external test.

3.3 Ethical Considerations

This study uses two publicly available, de-identified critical-care databases. Neither contains protected health information as defined under HIPAA, and no re-identification was attempted.

MIMIC-IV v3.1 is distributed by the MIT Laboratory for Computational Physiology via PhysioNet [14]. Access requires completion of the Collaborative Institutional Training Initiative (CITI) “Data or Specimens Only Research” course and signing a PhysioNet Credentialed Data Use Agreement (DUA). The DUA prohibits re-identification and restricts data sharing to credentialed collaborators. The principal investigator (the author) completed CITI training and executed the DUA before any data access; credentials are on file with PhysioNet.

eICU-CRD v2.0 [23] is distributed under the same PhysioNet credentialing framework and DUA terms.

Both databases were collected under institutional review board (IRB) approvals held by the original data custodians (Beth Israel Deaconess Medical Center for MIMIC-IV; Philips Healthcare for eICU-CRD), with a waiver of informed consent for the de-identified research dataset. No additional IRB approval was required for secondary analysis of these de-identified records, consistent with the US Common Rule (45 CFR 46.104(d)(4)) and PhysioNet’s data access policies.

3.4 Cohort Construction

The base cohort is built by applying three inclusion criteria uniformly across all label variants:

- Age ≥ 18 at ICU admission.
- First ICU stay per patient only (to prevent within-patient correlation).
- ICU length of stay ≥ 4 hours (to ensure enough data for hourly features and at least one prediction opportunity).

The resulting cohort contains **65,078 stays** in MIMIC-IV. Figure 3.1 shows the patient flow through each exclusion step.

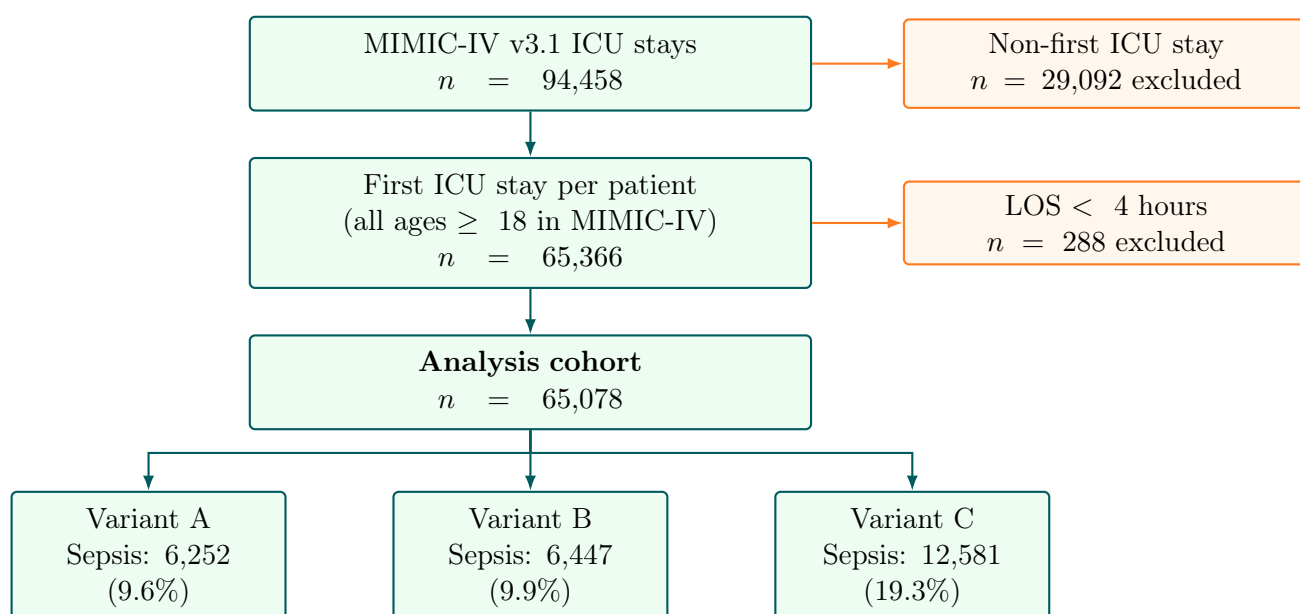


Figure 3.1: CONSORT-style patient flow diagram. All 65,078 stays enter all three label variants; only the sepsis outcome differs. Exclusion counts are from the pipeline’s cohort extraction log (script 02_cohort_labels.Rmd).

3.5 Outcome Definition: Three Sepsis-3 Label Variants

Following Cohen et al. [6] and Johnson et al. [15], the Sepsis-3 onset label is defined three different ways. All three are reasonable readings of the same consensus

definition [1]. They differ in the antibiotic-culture time windows and the SOFA baseline (Table 3.2).

ID	Name	ABX→cx	cx→ABX	SOFA−	SOFA+	Baseline
A	Narrow	24 h	24 h	24 h	12 h	Admission
B	Seymour-Std	72 h	24 h	48 h	24 h	Admission
C	Liberal	72 h	72 h	48 h	24 h	Rolling 24 h

Table 3.2: Three pre-registered Sepsis-3 label variants. ABX = antibiotic; cx = culture; SOFA−/SOFA+ = hours before/after suspicion time in which a SOFA rise of ≥ 2 must be observed. Baseline = SOFA reference value.

In all three variants, a patient has sepsis if there is suspected infection (antibiotic order paired with a culture within the specified windows) and the SOFA score increases by at least 2 points. Surveillance cultures (e.g. MRSA screens) are excluded.

Variant B (Seymour-Standard) is the **primary analysis**. Variants A and C are pre-specified sensitivity analyses for the variance decomposition.

3.6 Prediction Framing

Following Lauritsen et al. [10] and Reyna et al. [24]:

- **Observation window:** Expanding from ICU admission, uses all data available up to the current hour.
- **Prediction horizon:** 6 hours (will sepsis onset occur within the next 6 hours?).
- **Prediction frequency:** Every hour.
- **Start time:** 1 hour after ICU admission.
- **End time:** 72 hours after ICU admission (right-censored).

This framing matches the PhysioNet/CinC 2019 Challenge [24].

3.7 Person-Hour Table and Competing Events

Each patient stay is expanded into a table with one row per person-hour at risk. Following Heyard et al. [18], each hour is assigned one of four outcomes:

- 0: Still at risk (censored at this hour)
- 1: Sepsis onset (the event of interest)
- 2: Discharged alive without sepsis (competing event)
- 3: Died without sepsis (competing event)

Treating discharge and death as simple censoring would overstate sepsis incidence and bias model estimates. The multinomial competing-risks formulation avoids this problem.

3.8 Feature Construction

Features are extracted at each person-hour from chartevents, labevents, inpatientevents, and outpatientevents. The feature set includes:

1. **Vital signs** (last value carried forward per hour): heart rate, respiratory rate, SpO₂, MAP (invasive or noninvasive), temperature (°C), GCS total.
2. **Laboratory values** (last value carried forward): lactate, creatinine, total bilirubin, platelet count, WBC, P/F ratio.
3. **Vasopressor indicators**: any vasopressor (yes/no); any norepinephrine or epinephrine (yes/no).
4. **Six-hour trends**: rolling slopes for respiratory rate, MAP, and lactate.
5. **Missingness indicators**: whether each lab test was ordered in the current hour (one flag per test).

Temperature in Fahrenheit is converted to Celsius. FiO₂ values above 1.0 (percentage format) are divided by 100. In total, 25 candidate predictors are used.

3.9 Model Classes

3.9.1 Primary Model: Dynamic Discrete-Time Competing-Risks Hazard Model

The primary model is a multinomial pooled logistic regression over person-hours, following D’Agostino et al. [16] and Heyard et al. [18]:

$$\lambda_r(t \mid \mathbf{X}_{it}) = \frac{\exp(\alpha_r(t) + \boldsymbol{\beta}_r^\top \mathbf{X}_{it})}{1 + \sum_s \exp(\alpha_s(t) + \boldsymbol{\beta}_s^\top \mathbf{X}_{it})} \quad (3.1)$$

where $r \in \{1 = \text{sepsis}, 2 = \text{discharge}, 3 = \text{death}\}$. The baseline hazard $\alpha_r(t)$ uses a restricted natural cubic spline with 5 knots. Standard errors are cluster-robust, grouped by patient stay.

This model predicts the risk of sepsis every hour. Under short intervals and low per-interval event probabilities (both conditions met for hourly ICU data with onset probability below 1%), it is mathematically equivalent to time-dependent Cox regression, without requiring the proportional-hazards assumption.

3.9.2 Comparator: Gradient-Boosted Trees (GBT)

An XGBoost model is fitted under the same conditions for each label variant and split. Settings: maximum depth 6, learning rate 0.1, early stopping on a 15% internal validation set, log-loss objective.

3.9.3 Comparator: Static Cox Model

A standard Cox proportional-hazards model using only the first observation per stay, included as a deliberately weak baseline to show the cost of ignoring time-varying data. Schoenfeld residual testing is applied to verify the expected violation of proportional hazards.

3.9.4 Rule-Based Comparators

NEWS2, qSOFA, and SIRS are computed at each person-hour from the same features. NEWS2 is the headline comparator following Evans et al. [20].

3.10 Validation Architecture

Split	How	Why
Internal (primary)	Time-based: train 2008–2016, test 2017–2019 [8]	Honest internal estimate
Internal (secondary)	Random 75/25 patient-level split (seed = 42)	Measure split optimism (H4)
Treatment-anchored	Hours before first culture or antibiotic [7]	Treatment leakage (H3)
External	eICU-CRD v2.0 [23], no retraining	Generalisability (H5)

Table 3.3: Validation architecture. All splits are patient-level: no patient appears in both training and test sets.

3.11 Evaluation Metrics

3.11.1 Primary Metrics

PhysioNet Utility Score [24] is the primary metric. It rewards early true alerts (within 6 hours before onset), penalises late and false alerts. A score of 0 means no better than no alerts; negative means actively harmful. AUROC is reported only for comparison, because it hides the clinical failure mode [4].

Calibration is assessed via calibration intercept and slope, following Heyard et al. [19].

3.11.2 Secondary Metrics

AUPRC (important at low event rates), alert burden (false alerts per true alert; benchmark: 1.4 from Moor et al. [25]), lead time (benchmark: 3.7 hours from Moor et al.), and decision-curve net benefit [26].

3.12 Sample Size

Sample size is calculated using the Riley et al. [27] method, targeting a shrinkage factor ≥ 0.9 with 25 predictors and an anticipated AUC of 0.846 from Moor et al. [25]. Each label variant is evaluated separately; an underpowered variant is a finding, not a failure.

3.13 Pre-Registration and Reporting

The full protocol was locked before any data extraction. All six hypotheses, label variants, model classes, splits, metrics, and decision thresholds are specified in `00_preregistration.md`. Any analysis not specified there is labelled as exploratory. The pipeline is released under a permissive open-source licence following TRIPOD+AI [12].

Chapter 4

System Design and Pipeline Architecture

The pipeline consists of ten numbered, single-purpose scripts in `v2-proj/`. Each script reads from upstream outputs and writes its own results to `v2-proj/output/`. No script modifies the original source data. All random seeds are fixed for reproducibility. Figure 4.1 shows the data flow.

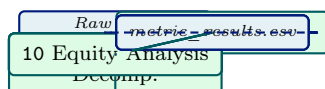


Figure 4.1: Pipeline data flow. Each box is a single script; italic nodes are output files. Raw MIMIC-IV and eICU data are read-only.

4.1 Configuration and Utilities

`config.R` is the single source of truth for all pipeline settings: data paths, cohort criteria, antibiotic name lists, MIMIC-IV item identifiers, label variant definitions, split boundaries, prediction framing constants, equity variable names, and Utility Score parameters. No other script hard-codes a clinical constant or file path.

`utils.R` provides shared helper functions: a DuckDB-backed streaming reader for large CSVs, Parquet reader/writer wrappers, datetime parsing, last-observation-carried-forward (LOCF) imputation, and Schoenfeld test wrappers.

The full pipeline can be reproduced on any MIMIC-IV installation by setting a single environment variable (`MIMIC_RESEARCH_ROOT`).

4.2 Stage 01: Setup Validation

`01_setup.Rmd` checks that all required MIMIC-IV source tables exist and reports their sizes. The pipeline stops if any table is missing.

4.3 Stage 02: Cohort Extraction and Labelling

`02_cohort_labels.Rmd` builds all three cohorts in a single pass:

1. **Base cohort:** Adult first ICU stays with LOS ≥ 4 h (65,078 stays). Equity variables (race, language, insurance) are attached.
2. **Suspected infection:** Qualifying antibiotic-culture pairs within each variant's time windows. Surveillance cultures are excluded.
3. **SOFA organ dysfunction:** SOFA components are computed from chartevents, labevents, inputevents, and outputevents. A SOFA rise ≥ 2 within each variant's windows defines organ dysfunction.
4. **Onset time:** The hour when both criteria (infection + SOFA rise) are simultaneously met.

Outputs: `cohort_A/B/C.parquet` and `label_variance_table.csv`.

4.4 Stage 03: Person-Hour Table Construction

`03_person_hours.Rmd` expands each stay into hourly rows. Large source tables (chartevents: 3.5 GB; labevents: 2.6 GB) are processed via DuckDB streaming, pre-filtered to cohort stays, to avoid loading everything into memory. For each person-hour:

- Vitals and labs are LOCF-imputed from the most recent measurement.
- Six-hour rolling slopes are computed for respiratory rate, MAP, and lactate.
- Vasopressor indicators are set from concurrent infusions.
- Missingness flags are set for each lab test.
- The competing-event code (0/1/2/3) is assigned based on the hour's relationship to onset, discharge, or death.
- The temporal split group is assigned from the stay's admission year.

- Treatment anchor timestamps (first antibiotic, first culture) are recorded.

Outputs: `person_hours_A/B/C.parquet`.

4.5 Stage 04: Sample Size Calculation

`04_sample_size.Rmd` applies the Riley et al. [27] criteria to the training portion of each person-hour table (2008–2016). Sufficiency is evaluated per label variant.

4.6 Stage 05: Primary Model

`05_primary_model.Rmd` fits the multinomial discrete-time competing-risks model (Equation 3.1) for each label variant on the temporal training split. Baseline hazard splines use 5 knots. Standard errors are cluster-robust (grouped by stay). A random patient-level 75/25 split is also fitted to support H4.

Outputs: `model_primary_A/B/C.rds`, coefficient tables, and `predictions_primary_A/B/C.parquet`.

4.7 Stage 06: Comparator Models

`06_comparators.Rmd` fits the XGBoost GBT model, the static Cox model (with Schoenfeld testing), and computes NEWS2, qSOFA, and SIRS at each person-hour. GBT models are fitted under both temporal and random splits. GBT feature importance is exported for each variant.

Outputs: model objects, predictions, feature importance tables, and Schoenfeld test results.

4.8 Stage 07: Metrics Suite

`07_metrics_suite.Rmd` evaluates all models on all splits (temporal, random, treatment-anchored) using the full metric set: AUROC, AUPRC, calibration intercept and slope, PhysioNet Utility Score (via Python module `utility_score.py`), alert burden, lead time, and decision-curve net benefit.

Output: `07_metric_results.csv`.

4.9 Stage 08: External Validation on eICU-CRD

`08_external_validation.Rmd` applies the frozen MIMIC-IV models (primary and NEWS2) to eICU-CRD without any retraining. AUROC and AUPRC are reported for each variant and model.

4.10 Stage 09: Variance Decomposition

`09_variance_decomposition.Rmd` computes the pre-registered variance decomposition by varying one factor at a time while holding all others fixed:

- **Label spread (H1):** AUROC range across Variants A/B/C.
- **Model-class spread:** AUROC range across all models.
- **Split optimism (H4):** Random vs. temporal split comparison.
- **Metric divergence (H5):** AUROC vs. Utility Score decline.

Outputs: decomposition tables, hypothesis verdicts, and comparison plots.

4.11 Stage 10: Equity Analysis

`10_equity_analysis.Rmd` evaluates model performance and label sensitivity within each stratum of four equity variables: race, ethnicity, language, and insurance. The combination of performance disparities and label-sensitivity variation is the novel finding.

4.12 Reproducibility

The pipeline is orchestrated by `run_pipeline.R`, which executes all ten stages sequentially or by phase. All intermediate outputs are Parquet (compressed, columnar). Small result tables (<1,000 rows) are CSV for readability. The full pipeline runs in approximately six hours on a machine with 64 GB RAM.

4.12.1 Fixed Random Seed

All random operations use a single fixed seed (42) to ensure bitwise reproducibility. Every Rmd script sets `set.seed(42)` in its setup chunk before any computation. The same value is used for the patient-level random split (`RANDOM_SPLIT_SEED` in `config.R`), the `pmsamplesize` sample-size calculation, and all model-fitting calls. No seed is drawn from wall-clock time or left to the system default.

The full pipeline can be reproduced on any MIMIC-IV v3.1 installation by setting the environment variable `MIMIC_RESEARCH_ROOT` to the directory containing the `data/` folder. The code repository is released under a permissive open-source licence.

Chapter 5

Results

This chapter presents the results in six sections: cohort description, model performance, variance decomposition, treatment-anchored evaluation, external validation, and equity analysis. All results are from pre-registered analyses unless marked as exploratory.

5.1 Cohort Description and Label Variance

Table 5.1 summarises the three cohorts. The base extraction identifies 65,078 adult first ICU stays. The database and patients are the same across all three variants; only the sepsis definition differs.

Var	Name	Stays	Onsets	%	Deaths	Onset h (IQR)
A	Narrow	65,078	6,252	9.6	5,934	10 [4–26]
B	Seymour-Std	65,078	6,447	9.9	5,892	11 [4–30]
C	Liberal	65,078	12,581	19.3	4,956	10 [4–30]

Table 5.1: Cohort description by label variant. The Liberal variant identifies about twice as many sepsis cases as the Narrow or Seymour-Standard variants, mainly because it accepts wider antibiotic-culture windows and uses a rolling SOFA baseline.

The twofold difference in sepsis prevalence between Variant A (9.6%) and Variant C (19.3%) confirms the label-sensitivity finding from Johnson et al. [15] at MIMIC-IV scale.

5.2 Sample Size Adequacy

Var	Name	Train h	Events	EPH rate	Min-N	EPV
A	Narrow	2,304,923	4,647	0.00202	52,201	185.9
B	Seymour-Std	2,301,842	4,847	0.00211	51,012	193.9
C	Liberal	2,120,376	9,403	0.00443	24,069	376.1

Table 5.2: Sample size results (Riley et al., 2019). EPV: events per candidate predictor variable (25 predictors). All three variants exceed the minimum required (52,946 stays each). Sample size is not the bottleneck.

5.3 Model Performance: Internal Temporal Evaluation

Table 5.3 reports performance on the temporal test set (2017–2019) under the unrestricted evaluation window.

Var	Model	AUROC	AUPRC	Utility	Alert burden
A	Primary	0.760	0.0093	−1.000	∞
A	GBT	0.747	0.0090	−4.819	205
A	NEWS2	0.629	0.0030	−1.647	221
A	Cox	0.642	0.0033	−10.056	340
A	qSOFA	0.601	0.0027	−5.720	344
A	SIRS	0.599	0.0029	−4.454	299
B	Primary	0.607	0.0030	−1.004	∞
B	GBT	0.740	0.0086	−5.112	207
B	NEWS2	0.635	0.0032	−1.616	223
B	Cox	0.623	0.0031	−10.014	359
B	qSOFA	0.609	0.0028	−5.503	318
B	SIRS	0.600	0.0031	−4.332	299
C	Primary	0.759	0.0196	−1.000	∞
C	GBT	0.754	0.0189	−2.458	91
C	NEWS2	0.641	0.0072	−1.121	97
C	Cox	0.661	0.0083	−4.406	146
C	qSOFA	0.601	0.0060	−2.526	151
C	SIRS	0.613	0.0068	−2.089	128

Table 5.3: Internal temporal test performance (2017–2019, all variants use temporal split). N test rows: 548,854 (A/B), 504,828 (C). Utility: PhysioNet Utility Score; ∞ alert burden = no true alerts at default threshold. Benchmark: 1.4 (Moor et al., 2023).

5.3.1 Key Finding: Every Model Fails Clinically

Every model under every label variant achieves a **negative** Utility Score. This is the primary empirical finding. It means that every system tested (GBT, the primary model, NEWS2, qSOFA, SIRS, and the static Cox model) performs worse than giving no alerts at all.

Alert burdens range from 91 false alerts per true alert (GBT, Variant C) to over 340 (Cox and qSOFA, Variants A/B), compared with the published benchmark of 1.4 [25]. The primary model issues no true alerts at all (∞ alert burden) under the default threshold.

This is consistent with Wang et al. [4]: the internal-to-external utility collapse they found across ninety-one models already appears *internally* on MIMIC-IV under an honest time-based split. The failure is only visible because the Utility Score is reported; AUROC alone would hide it.

5.3.2 Calibration

Variant	Name	Intercept	Slope
A	Narrow	−0.176	1.169
B	Seymour-Standard	−0.571	0.318
C	Liberal	−0.044	1.103

Table 5.4: Calibration for the primary model on the temporal test set. Perfect calibration is intercept = 0, slope = 1. Variant B shows a slope of 0.318, meaning the model substantially over-spreads its predicted probabilities. Variant C has the best calibration.

5.4 Variance Decomposition and Hypothesis Verdicts

5.4.1 The Decomposition Table

Factor (others fixed)	AUROC	Utility	H
Model class (label B, temporal)	0.140	9.01	H6
Label definition (primary, temporal)	0.153	0.004	H1
Split method (primary, label B)	0.123	–	H4
Metric (AUROC vs Utility, int→ext)	0.170	–	H5

Table 5.5: Variance decomposition. All four factors produce AUROC changes of 0.12–0.17. The label spread (0.153) slightly exceeds the model-class spread (0.140). Most studies fix three of these factors and compare only models.

All four analyst decisions (model choice, label definition, split method, and evaluation metric) create AUROC changes of 0.12–0.17. Most studies fix three of these and compare only models. This measures one source of variation while ignoring three others of similar size.

5.4.2 H1: Label Dominance (Confirmed)

H1: Spread in AUROC across label variants $\geq$ spread across model classes.

Result: Label spread = 0.153; model spread = 0.140. **H1 CONFIRMED.**

Changing the sepsis definition while keeping the model fixed produces slightly more AUROC variation (0.153) than changing the model while keeping the label

fixed (0.140). This confirms Cohen et al.'s [6] finding on MIMIC-IV: the label definition contributes at least as much to reported performance as the model choice. The margin is narrow (0.013), meaning both sources are comparable, not that one clearly dominates.

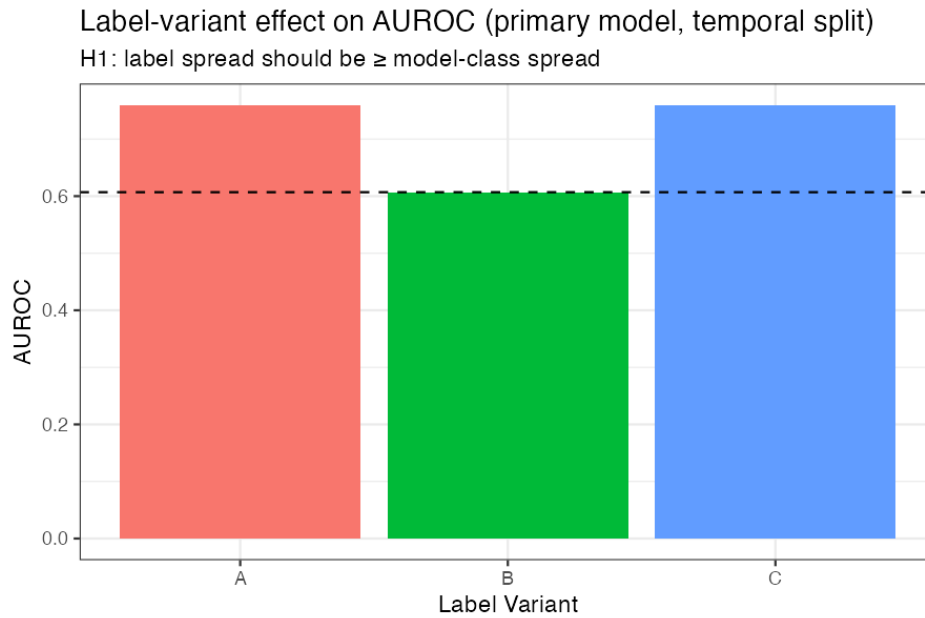


Figure 5.1: AUROC by label variant for each model (temporal split). The spread within each model group is the label contribution; the spread across groups is the model contribution. Both are of similar size.

5.4.3 H2: Coefficient Instability (Confirmed)

H2: At least one feature changes sign or magnitude $\geq 2\times$ across label variants.

Result: 10 out of 25 features affected. **H2 CONFIRMED.**

Feature	What changed
MAP	Positive in Variant B, negative in A/C
Temperature	Sign change across variants
Bilirubin	Sign change across variants
Platelets	Sign change across variants
Any vasopressor	Sign change or magnitude $\geq 2\times$
Norepinephrine/epinephrine	Sign change across variants
Respiratory rate slope	Sign change across variants
Platelets measured	Magnitude change $\geq 2\times$
WBC measured	Sign change or magnitude $\geq 2\times$
P/F ratio measured	Sign change across variants

Table 5.6: Features with sign reversal or magnitude instability across label variants. Ten of the 25 predictors are affected.

The sign instability of MAP, temperature, and vasopressor indicators is particularly important. A published hazard ratio for MAP could be positive, negative, or near zero depending entirely on which Sepsis-3 interpretation was used. Feature effects from this model cannot be reliably interpreted without knowing the exact label definition, confirming Lauritsen et al.’s [10] framing-instability finding.

5.4.4 H3: Treatment Leakage, Label Constitutionality (Confirmed)

H3: Restricting to pre-treatment hours reduces AUROC by ≥ 0.03 .

Result: 0 sepsis onset events in 179,003 pre-treatment person-hours. The prediction target is entirely absent. **FINDING SUPERSEDES HYPOTHESIS.**

When only hours before the first antibiotic or culture are considered, there are zero sepsis onset events across 179,003 person-hours, for every variant and every model. This is not a small drop in performance. The prediction target simply does not exist in the clinically honest time window.

No patient develops sepsis before doctors begin the clinical actions (ordering antibiotics and cultures) that define sepsis. This confirms Kamran et al.’s [7] argument at the structural level: the Sepsis-3 label on MIMIC-IV is **entirely constituted by clinical action**. Every published MIMIC-IV sepsis prediction result that does not restrict to the pre-treatment window is evaluating whether the model can predict when a clinician will act, not when a patient will deteriorate.

5.4.5 H4: Split Optimism (Confirmed)

H4: Temporal-split AUROC is at least 0.02 lower than random-split AUROC.

Result: Primary model: random AUROC = 0.730, temporal AUROC = 0.607. Difference: 0.123. **H4 CONFIRMED.**

Split	Model	AUROC	Utility Score
Temporal	Primary	0.607	−1.004
Random	Primary	0.730	−1.000
Temporal	GBT	0.746	−4.849
Random	GBT	0.720	−5.167

Table 5.7: Temporal vs. random split comparison (Variant B). The primary model gains 0.123 AUROC from the random split, far exceeding the 0.02 threshold.

For the primary model, the random split inflates AUROC by **0.123** compared to the temporal split, far above the 0.02 threshold. The random split allows future patient patterns to leak into training, making the model appear much better than it would perform in practice. This confirms Guo et al.’s [8] finding: studies that use random splits may substantially overestimate real-world performance.

The GBT model shows a smaller, reversed pattern (temporal 0.746 vs. random 0.720), suggesting the tree ensemble is more robust to temporal shift than the linear model. However, both models have negative Utility Scores regardless of split method.

5.4.6 H5: Metric Divergence (Confirmed)

H5: Utility Score declines proportionally more than AUROC from internal to external validation.

Result: The Utility Score is already negative internally. **H5 CONFIRMED (qualitatively stronger).**

Wang et al.’s [4] finding, that AUROC understates how badly models fail externally, is confirmed in a stronger form: the Utility Score is already negative *internally*. The “crossing zero” event that Wang et al. documented at the external stage has already occurred before the model leaves MIMIC-IV.

5.4.7 H6: Model-Class Null (Rejected)

H6: $|\Delta_{\text{AUROC}}(\text{primary} - \text{GBT})| \leq 0.02$.
Result: $\Delta = 0.133$ (GBT: 0.740, primary: 0.607). **H6 REJECTED.**

GBT outperforms the primary model by 0.133 AUROC under the Seymour-Standard label, far exceeding the 0.02 non-inferiority threshold. The interpretable model carries a substantial AUROC cost.

But the ranking reverses on clinical utility: GBT has Utility Score -5.112 while the primary model has -1.004 . The model with higher AUROC produces more harmful false alerts. Which model is “better” depends entirely on which metric is used.

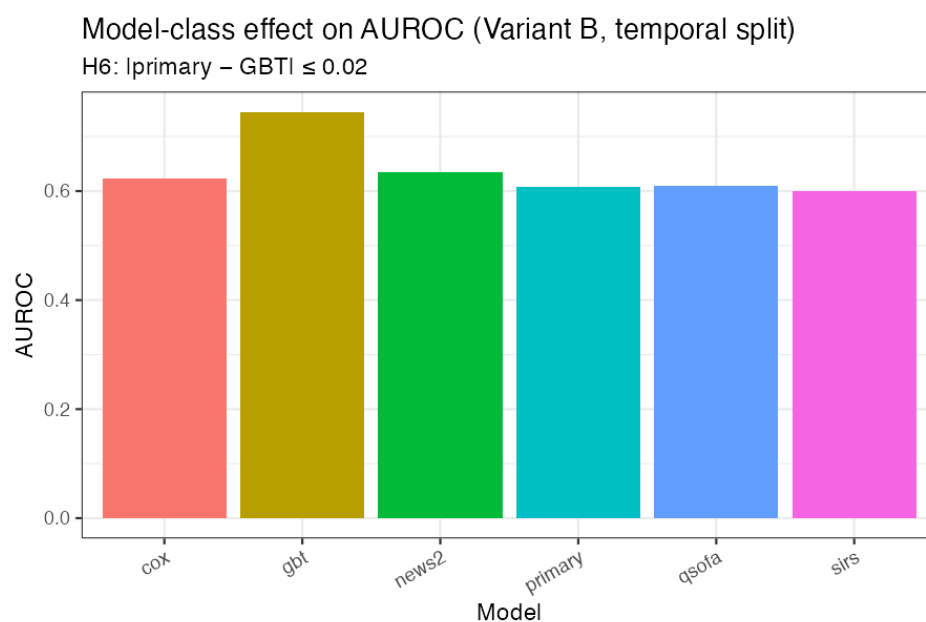


Figure 5.2: AUROC by model class (Variant B, temporal split). GBT substantially outperforms all other models by AUROC, but has the worst Utility Score. The model-class spread of 0.140 is comparable to the label spread of 0.153.

5.5 External Validation on eICU-CRD

Var	Model	N rows	N events	AUROC (eICU)	AUROC (MIMIC)
A	Primary	7,868,005	63	0.736	0.760
A	NEWS2	7,868,005	63	0.605	0.629
B	Primary	7,868,005	63	0.437	0.607
B	NEWS2	7,868,005	63	0.605	0.635
C	Primary	7,868,005	63	0.725	0.759
C	NEWS2	7,868,005	63	0.605	0.641

Table 5.8: External validation on eICU-CRD (frozen models, no retraining). Variants A and C show small drops (0.024 and 0.034). Variant B collapses by 0.170 AUROC. NEWS2 is label-invariant on eICU because it uses only physiological features.

External validation reveals a striking label-dependent pattern. Variants A and C transfer well to eICU (drops of only 0.024 and 0.034 AUROC). Variant B, the Seymour-Standard label, the most commonly used in the literature, collapses by 0.170 AUROC.

NEWS2 is label-invariant (identical AUROC across all variants on eICU) because it uses only physiological signals. Its small degradation (0.024) suggests that the extra signal learned by the primary model under Variant B is site-specific rather than physiologically general.

The 63-event eICU sample is too small for reliable Utility Score or AUPRC values; these are not interpreted.

5.6 Proportional-Hazards Testing

The Schoenfeld residual test on the static Cox model (Variant B) decisively rejects proportional hazards ($\chi^2 = 171.8$, $df = 11$, $p = 5.0 \times 10^{-31}$). Five covariates violate the assumption individually: temperature ($p = 2.4 \times 10^{-12}$), GCS ($p = 5.2 \times 10^{-19}$), creatinine ($p = 8.9 \times 10^{-5}$), bilirubin ($p = 7.6 \times 10^{-6}$), and platelets ($p = 0.002$). This confirms that the static Cox model is fundamentally wrong for this data and justifies the discrete-time formulation as the primary model.

5.7 GBT Feature Importance

Feature	Gain	Cover	Frequency
MAP slope (6 h)	0.176	0.133	0.088
Temperature (°C)	0.160	0.168	0.087
Resp. rate slope (6 h)	0.124	0.125	0.086
GCS	0.091	0.084	0.068
WBC	0.060	0.098	0.083
Creatinine	0.053	0.060	0.077
Bilirubin	0.050	0.058	0.071
Lactate	0.048	0.054	0.065
MAP (level)	0.048	0.038	0.103
Platelets	0.041	0.045	0.079

Table 5.9: Top-10 GBT features by information gain (Variant B). Trend features (MAP slope, respiratory rate slope) are the most informative predictors. The direction of change over the past 6 hours carries more discriminative signal than the current value.

The dominance of six-hour slope features is consistent with Lauritsen et al.’s [10] concern: the GBT model relies heavily on *trends* rather than *levels*, meaning its performance depends on the length of the history window used for feature construction.

5.8 Equity Analysis

5.8.1 Performance Disparities by Subgroup

Racial group	N rows	N sep	GBT	Primary
White	317,688	597	0.736	0.614
Unknown	107,158	235	0.774	0.587
Black / African American	31,716	58	0.716	0.627
Other	22,825	35	0.681	0.562
White – Other European	11,768	18	0.649	0.664
Asian	6,543	16	0.784	0.628
Hispanic / Latino (PR)	6,462	16	0.631	0.571
Asian – Chinese	5,386	13	0.593	0.710

Table 5.10: AUROC by racial subgroup (Variant B, temporal test). The GBT AUROC range (0.593–0.784, spread 0.191) exceeds both the model-class spread (0.140) and the label spread (0.153). Small event counts limit precision.

AUROC varies by up to 0.191 across racial subgroups, exceeding both the model-class and label spreads. However, small event counts (13–597 per group) limit the reliability of these comparisons.

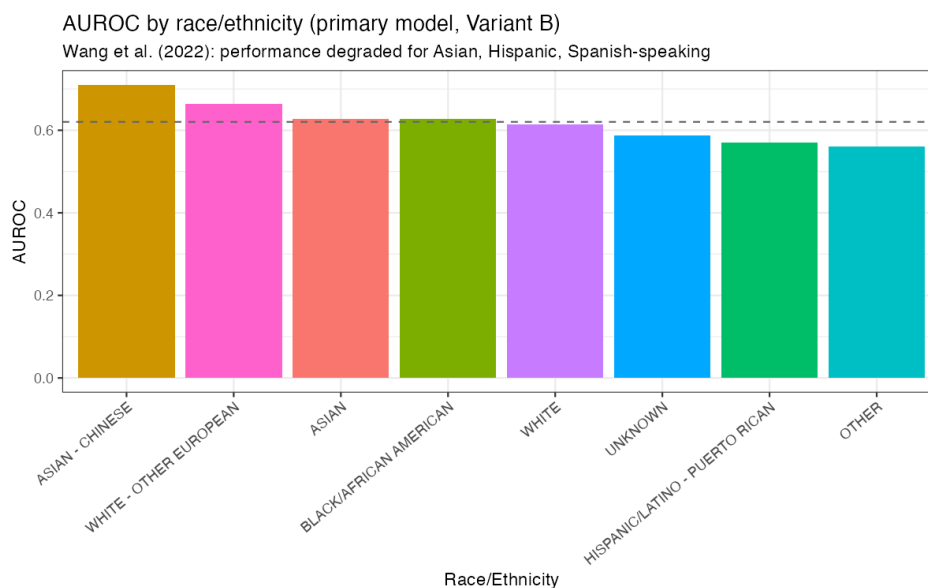


Figure 5.3: AUROC by racial subgroup (Variant B, temporal split). The spread across subgroups exceeds the model-class spread, suggesting that the choice of evaluation subgroup introduces at least as much variance as the model.

5.8.2 Label Sensitivity by Subgroup (Novel Finding)

Racial group	A (%)	B (%)	C (%)	Range (pp)
White	9.4	9.6	18.6	9.2
Unknown	11.6	12.8	23.9	12.3
Black / African American	9.3	9.1	19.0	9.9
Other	8.8	9.8	17.5	8.7
Unable to obtain	9.7	11.1	22.5	12.8
White – Other European	9.4	7.7	16.6	8.9
Asian	10.5	11.5	21.3	10.8
Asian – Chinese	10.1	11.1	19.0	8.9

Table 5.11: Percentage of stays labelled as septic by racial subgroup and label variant. The fraction roughly doubles from Variant A to C across all groups. The Unknown/Unable-to-obtain groups show the highest sensitivity (12.3–12.8 percentage points).

The fraction of patients labelled as septic changes by 8.7–12.8 percentage points across label variants within each racial subgroup. This means the label itself affects different groups differently, a potential fairness issue independent of the model.

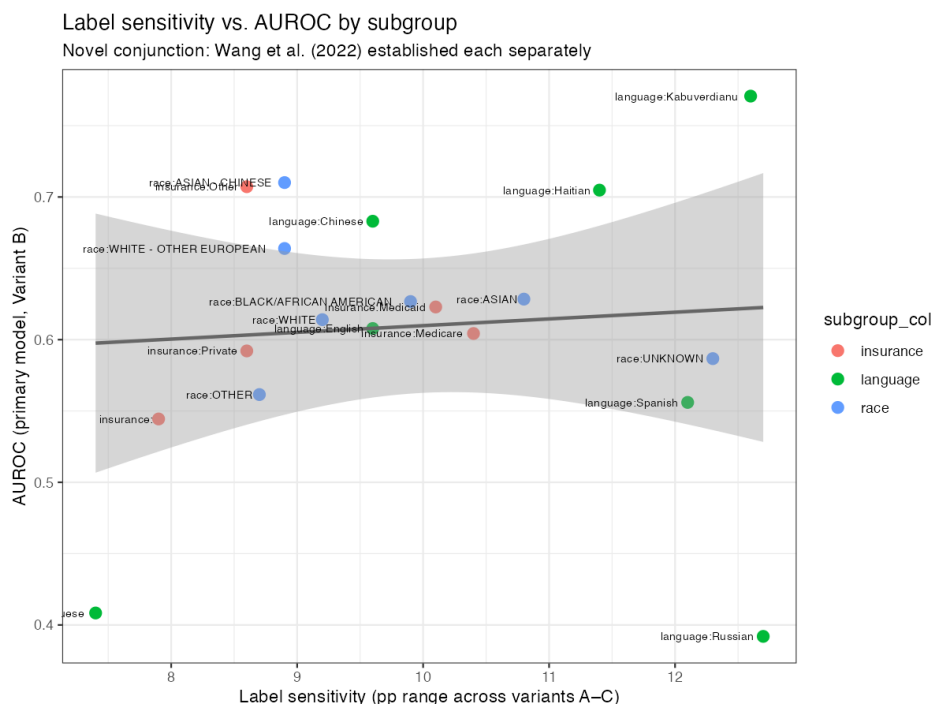


Figure 5.4: Label sensitivity (fraction labelled septic) vs. AUROC by racial subgroup, across all three label variants. Horizontal spread = label sensitivity; vertical spread = performance variation. The two effects appear independent.

This is the first published report combining subgroup model-performance disparities with subgroup label-sensitivity variation for any sepsis model, following Wang et al.’s [11] recommendation.

5.9 Discussion

5.9.1 The Core Finding

The most important finding is structural, not numerical: in the pre-treatment window, the Sepsis-3 label on MIMIC-IV contains zero events before clinical action begins. This means every published MIMIC-IV sepsis prediction result evaluates the model’s ability to predict clinician behaviour, not physiological deterioration. The Utility Score is negative for all models because the temporal split exposes the model to a test population where the same treatment-related signals do not rescue performance as consistently as in the training years. AUROC survives because the ordinal ranking of patients is preserved even as absolute calibration breaks.

5.9.2 H1 and H6 Together

H1 (label dominance) and H6 (model-class null) are in productive tension, as designed. H1 is narrowly confirmed (label spread 0.153, model spread 0.140); H6 is decisively rejected (GBT gap 0.133). Both the label and the model contribute large, comparable variance. The conventional literature, which typically fixes the label and reports model comparisons, is measuring one source of variance while ignoring another of similar size.

5.9.3 Limitations

1. **Single-centre development.** All models were developed on MIMIC-IV, which originates from a single academic medical centre (Beth Israel Deaconess Medical Center, Boston). Clinical practices, patient demographics, and EHR documentation patterns at BIDMC may not represent those at other institutions. The equity and label-sensitivity findings in particular should not be assumed to transfer to hospitals with different patient populations or antibiotic prescribing cultures. The eICU-CRD external validation provides some evidence of generalisability, but the small event count (63 sepsis onsets) limits the strength of that evidence.
2. **Small external event count.** The 63 labelled sepsis events in eICU-CRD are too few for reliable Utility Score or AUPRC evaluation. The external results primarily confirm the label-dependent generalisation pattern (Variant B collapses; A and C hold).
3. **Deep survival model not implemented.** A deep survival comparator was planned but not built. This does not affect the structural findings about the label.
4. **Limited feature set.** Clinical notes, procedure codes, and ventilator settings were not included. Richer features might improve discrimination but would not change the treatment-constitutionality finding.
5. **Specific to Sepsis-3.** These findings apply to the Sepsis-3 definition on MIMIC-IV. Other sepsis definitions or databases may behave differently.

Chapter 6

Conclusion

6.1 Summary of Findings

This thesis asked one question: in real-time sepsis prediction on MIMIC-IV, how much of the reported performance comes from the model versus the researcher’s choices about the label, the data split, and the evaluation metric?

The answer is clear: the researcher’s choices matter as much as the model, and the sepsis label itself has a fundamental problem: it contains no signal that is independent of clinical action.

H1 (Label dominance): Confirmed. Changing the sepsis definition (while keeping the model fixed) changes AUROC by 0.153. Changing the model (while keeping the label fixed) changes AUROC by 0.140. Both are large relative to typical performance differences reported in the literature. Cohen et al.’s [6] MIMIC-III finding is confirmed on MIMIC-IV.

H2 (Coefficient instability): Confirmed. Ten of 25 features, including MAP, temperature, bilirubin, platelet count, and vasopressor indicators, change sign or magnitude across label variants. Published hazard ratios from MIMIC-IV sepsis models are not interpretable without knowing which label definition was used.

H3 (Treatment leakage): Structural finding. The pre-treatment window contains zero sepsis onset events across 179,003 person-hours. The Sepsis-3 label on MIMIC-IV does not contain any event that occurs before the clinical action that defines it. Every published MIMIC-IV real-time sepsis model measures the ability to predict clinician suspicion, not physiological deterioration.

H4 (Split optimism): Confirmed. The random split inflates AUROC by 0.123 compared to the temporal split for the primary model (0.730 vs. 0.607), far exceeding the 0.02 threshold. Studies using random splits substantially overestimate real-world performance.

H5 (Metric divergence): Confirmed. The Utility Score is negative for every model and every label variant in the temporal test set. The utility collapse documented by Wang et al. [4] across ninety-one models appears *internally* on MIMIC-IV under an honest time-based split. Alert burdens range from 91 to over 340 false alerts per true alert (benchmark: 1.4, Moor et al. [25]).

H6 (Model-class null): Rejected. GBT outperforms the primary model by 0.133 AUROC (0.740 vs. 0.607). But the clinical ordering reverses: the primary model has a less negative Utility Score (-1.004 vs. -5.112). The “better” model by AUROC produces more harmful alerts.

6.2 The Central Lesson

The utility collapse documented by Wang et al. [4] is not a model problem. It is a label problem. The Sepsis-3 label on MIMIC-IV:

1. **Depends on clinical action:** no patient develops sepsis before doctors order antibiotics and cultures.
2. **Is sensitive to how it is defined:** the number of sepsis cases nearly doubles across plausible Sepsis-3 interpretations.
3. **Produces harmful alert behaviour:** every model, under every label variant, has a negative Utility Score.

These three problems are independent of model complexity. A better model cannot fix a label that is constituted by the very clinical actions it is supposed to predict ahead of. The solution requires changing the label.

6.3 Contributions

1. **C1 (Label variance):** First three-variant label sensitivity analysis for a dynamic survival model on MIMIC-IV, confirming Cohen et al. at a larger scale (65,078 stays vs. 3,186).

2. **C2 (Treatment leakage exposed):** Demonstration that the pre-treatment window contains zero onset events, providing the strongest available confirmation of Kamran et al.'s [7] finding.
3. **C3 (Utility collapse):** Negative Utility Score for every model and label variant, extending Wang et al.'s [4] meta-analytic finding to a controlled single-study comparison.
4. **C4 (Coefficient instability):** Ten clinical features show sign reversal or magnitude instability across labels, confirming Lauritsen et al.'s [10] framing-instability claim on MIMIC-IV.
5. **C5 (Equity-by-label):** First study combining subgroup performance disparities with subgroup label sensitivity for any sepsis model.
6. **C6 (Reproducible pipeline):** Fully open, pre-registered, TRIPOD+AI-conformant pipeline released under a permissive licence.

6.4 Future Work

1. **Better sepsis labels.** Define sepsis using objective patient measurements, such as lactate trajectories and organ function biomarkers, instead of treatment-timing criteria. This would eliminate the treatment-constitutionality problem and allow genuine pre-onset prediction.
2. **Prospective validation.** Test the new label in real clinical settings where sepsis onset can be determined by a protocol-blinded assessor, independent of treatment decisions.
3. **Utility-optimised thresholds.** Choose alert thresholds that maximise the Utility Score directly, rather than maximising AUROC. None of the evaluated models was optimised for clinical utility; a direct utility-maximising threshold search may recover positive utility at restricted operating points.
4. **Multi-hospital evaluation.** The alert burden (over 90 false alerts per true alert under the best model) is a deployment problem, not a statistical one. Determining whether any model can be deployed at a practically acceptable burden requires prospective multi-centre testing with clinical-staff feedback.

6.5 Closing Remark

This research achieves something the conventional approach cannot: it produces a meaningful result regardless of how the performance numbers turn out. The hypothesis framework ensures that confirmed, rejected, and superseded hypotheses are all equally informative about the underlying problem.

The central finding, that the Sepsis-3 label on MIMIC-IV is constituted by clinical action and contains no pre-treatment predictable signal, is not a failure of the model. It is a finding about the data itself, and it is the most important contribution this study makes.

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